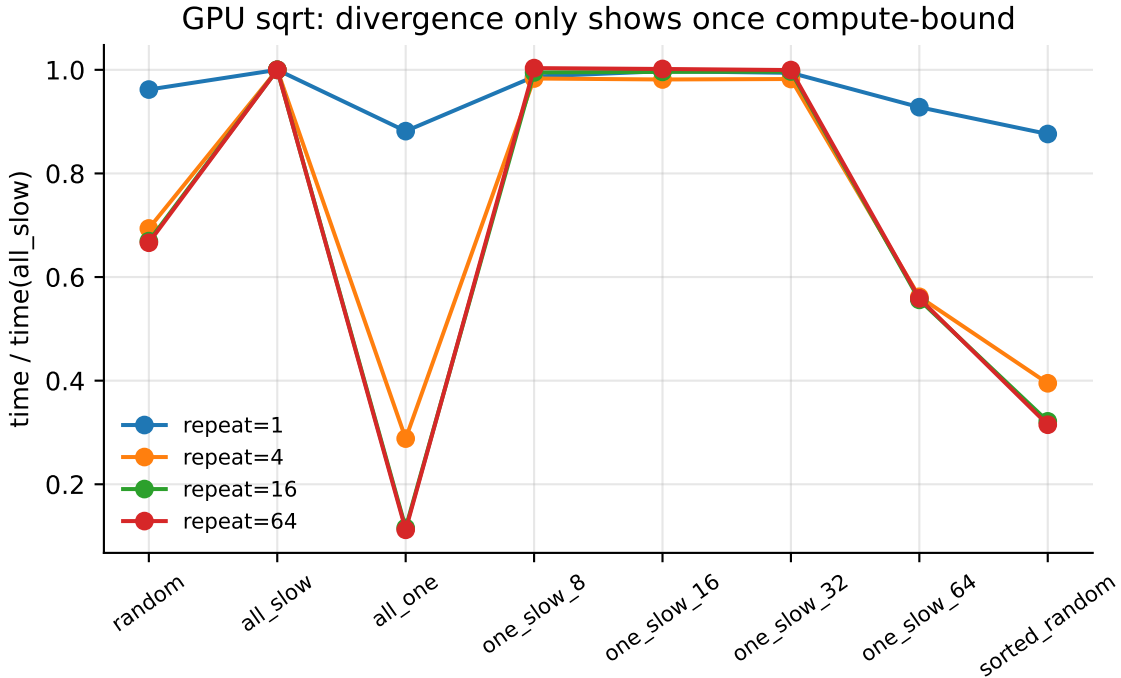
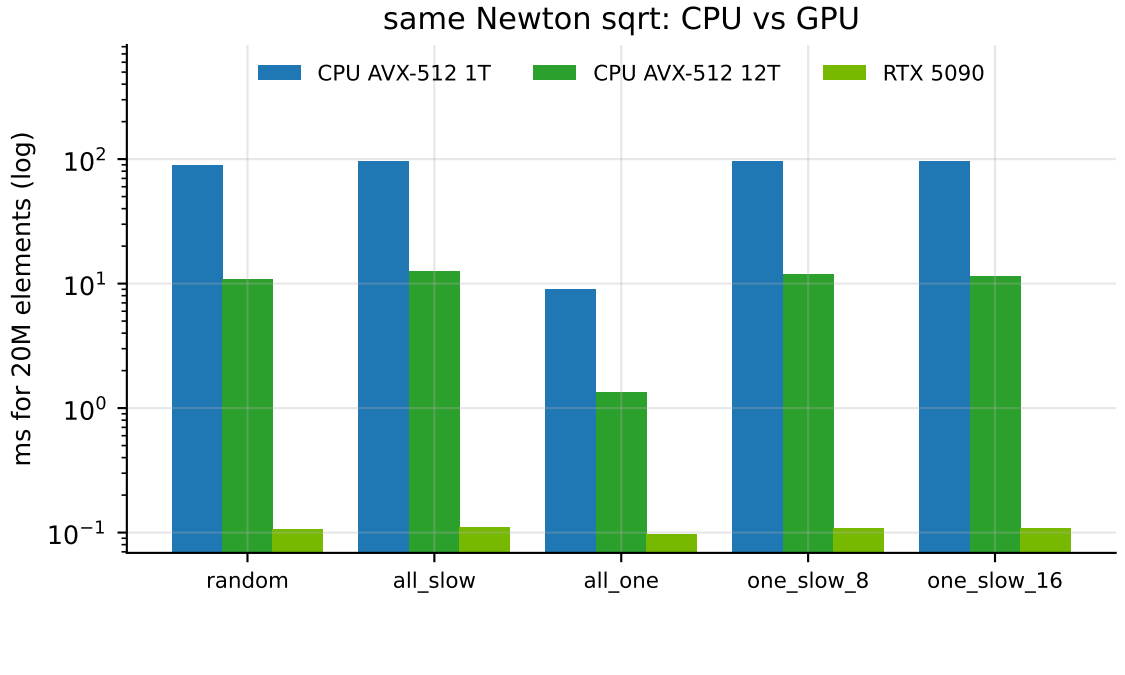

GPU sqrt: divergence only shows once compute-bound
time / time(all_slow)
1.0
0.8
0.6
0.4
0.2
repeat=1
repeat=4
repeat=16
repeat=64
random
all_slow
all_one
one_slow_8
one_slow_16
one_slow_32
one_slow_64
sorted_random


same Newton sqrt: CPU vs GPU
CPU AVX-512 1T
CPU AVX-512 12T
RTX 5090
ms for 20M elements (log)
10^2
10^1
10^0
10^{-1}
random
all_slow
all_one
one_slow_8
one_slow_16